

Department of Pharmacy
Choose the best alternative answer.

Cheating and chatting absolutely restricted

1. Requirements to register as pharmacy professional includes all Except.
 - A. Age
 - B. Registration board
 - C. Citizenship
 - D. None
2. Which of the following approach to stock management requires that the products will expire first is issued first?
 - A.LIFO
 - B.FIFO
 - C.FEFO
 - D.NONE
3. What are the reasons for conducting a drug use evaluation study?
 - a. To describe and compare current pattern of medicines use
 - b. To identify drug use problems
 - c. To monitor medication use over time
 - d. All
4. Which is an example of prescribing indicators?
 - a. Percent of encounters prescribed by generic
 - b. Percent of encounters with antibiotics prescribed
 - c. Percent of encounters with injections prescribed
 - d. All
 - e. None
5. During a tablet film coating, a supervising pharmacist observed tablets mottled forming uneven color distribution. What kind of measure should she taken to solve the problem?
 - A. Dispersing pigment aggregates
 - B. Using dyes instead of pigments
 - C. Decreasing drying temperature
 - D. Pouring coating solution rather than atomization
6. A 40 year old hypertensive patient while taking anti-hypertensive drugs suffered from a precipitous fall in blood pressure and fainting on initial administration. It is commonly seen with:
 - a. Nifedipine
 - b. Prazosin
 - c. Atenolol
 - d. Verapamil

7. In a patient of hypertension, the dose of propranolol that normalized blood pressure, reduced resting heart rate to 50/min. Which of the following β blockers will be most suitable for him as an alternative so that heart rate is not markedly reduced:
 - a. Pindolol
 - b. Labetalol
 - c. Bisoprolol
 - d. Atenolol
8. A 45 years old male takes simvastatin for hypercholesterolemia; however, his cholesterol level remains above target at maximal doses. Cholestyramine is added to the therapeutic regimen. What drug- Drug interactions can occur?
 - b. The combination will not lower cholesterol more than either agent alone
 - c. The combination causes elevated very low-density lipoprotein
 - d. Cholestyramine inhibits GI absorption of simvastatin
 - e. Simvastatin is a direct antagonist of Cholestyramine
9. A patient suffers from troublesome allergic rhinitis due to pollen, and you want to prescribe a drug for her that is least likely to cause sedation. What would your best choice be?
 - a. Betamethasone
 - b. Hydroxyzine
 - c. Loratadine
 - d. Metoclopramide
10. JC is an 8 year old boy who is brought to the Emergency Department by his parents after suffering from an asthmatic attack. Which of the following would be LEAST likely to reverse bronchoconstriction in this situation?
 - a. cromolyn sodium by nebulizer
 - b. epinephrines.c.
 - c. ipratropium aerosol
 - d. terbutaline, s.c.
 - e. theophylline i.v
11. A case of poisoning has been brought to the emergency department of a hospital with weakness, fainting, uncontrolled passage of urine and stools, profuse sweating, salivation, watering from nose and eyes. The most likely cause of poisoning:
 - A. Atropine
 - B. Phenylephrine
 - C. Malathion
 - D. None of the above
12. A 48 years old man presents with a blood pressure of 146/95mmHg. He is treated with a diuretic at a dose of 25mg once daily. While his blood pressure is under control, he later develops hypokalemia. Which of the following diuretics has been most likely prescribed to the patient?
 - a. Acetazolamide
 - b. Amiloride
 - c. Eplerenone
 - d. Hydrochlorothiazide
13. A 65-year-old male is brought to the emergency department due to shortness of breath. He is diagnosed with pulmonary edema due to acute decompensated heart failure. He is

- started on a diuretic to improve his congestion. The most likely diuretic for this condition?
- Chlorthalidone
 - Furosemide
 - Mannitol
 - Triamterene
14. A 60 year old male has been placed on a diuretic to control his high blood pressure. However, after a few weeks, he develops an acute gouty attack. If the gouty attack is precipitated by the diuretic therapy, which of the following drugs has he most likely been taking?
- Acetazolamide
 - Eplerenone
 - Hydrochlorothiazide
 - Torsemide
15. A 72 years old female with chronic systolic heart failure of 3 years of duration is being placed on spironolactone in addition to standard therapy to improve symptoms and prolong survival. What is the most likely mechanism of action for spironolactone to prolong this patient's life?
- Decreased cardiac remodeling
 - Increased levels of Potassium
 - Increased diuresis
 - Inhibition of Na^+/K^+ ATPase
 - Positive inotropic effect
16. A 60-year-old female with deep vein thrombosis is prescribed a drug that inhibits the enzyme that converts the inactive vitamin K 2,3-epoxide into the active, reduced form of vitamin K. Which of the following drugs is she most likely given?
- Aspirin
 - Warfarin
 - Prasugrel
 - Tinzaparin
17. A 65 years old male develops severe thrombocytopenia in response to treatment with an anticoagulant. Which of the following drugs is most likely responsible for this patient's thrombocytopenia?
- Dabigatran
 - Dipyridamole
 - Prasugrel
 - Unfractionated heparin
18. A 36-year-old man presents with tingling and neuropathic pain in his feet. Blood test reveals macrocytic anemia. Which of the following drugs will probably be required in this case?

- a. Erythropoietin
 - b. Filgrastim
 - c. Vitamin B12
 - d. Folic acid
- 19.** A 35-year-old man has severe schizophrenia. He is institutionalized and maintained on antipsychotic medication. Which of the following drugs is most likely to cause parkinsonian adverse effects?
- a. Clozapine
 - b. Risperidone
 - c. Haloperidol
 - d. Olanzapine
 - e. is usually started concurrently with phenobarbital therapy
- 20.** A 23-year-old woman with a history of seizure disorder is planning to become pregnant. She realizes that being on this medication during pregnancy can have fetal effects of mental retardation. Which of the following agents would likely produce a newborn with the lowest intelligence quotient(IQ)?
- a. Carbamazepine
 - b. Lamotrigine
 - c. Phenytoin
 - d. Valproic acid
- 21.** Drug X is an antimycobacterial agent that inhibits other bacteria as well as poxviruses. However, it should not be used as a single agent because resistant mutants frequently form. The responsible mutation may alter the site of action of drug X [i.e., the deoxyribonucleic acid (DNA)–dependent ribonucleic acid (RNA) polymerase]. What is drug X?
- a. INH
 - b. Rifampin
 - c. Pyrazinamide
 - d. Ethambutol

22. A 50-year-old male farm worker is brought delayed to the emergency room. He was found confused and since then has lost consciousness. His heart rate is 45, and his blood pressure is 80/40 mm Hg. He is sweating and salivating profusely. He has also diarrhea, Miosis and shortness of breath, Which of the following treatments is indicated?

- a. Atropine
- b. Adrenaline
- c. Bethanecol
- d. Pralidoxime
- e. methadone

23. A pregnant women with primary generalized tonic-clonic seizures, well-controlled on phenobarbital, stops taking her antiepileptic medications 4 months into her pregnancy. Which of the following describe her decision?

- A. Her decision is wrong, as the risk of teratogenicity was highest in the first trimester.
- B. Her decision is wrong because antiepileptic drugs do not increase the risk of fetal malformations
- C. Her decision is correct as the risk of seizures is reduced in pregnancy
- D. Her decision is wrong but her medication needs to be changed and a newer antiepileptic drug added.
- E. Her decision is correct as phenobarbital can cause vitamin K deficiency

24. A 50-year-patient with edema of the ankles develops gynecomastia and erectile dysfunction while being treated with which of the following agents?

- A. Hydrochlorothiazide
- B. Spironolactone
- C. Metolazone
- D. Amiloride

25. A female patient with a mechanical heart valve who is taking warfarin informs you that she hopes to get pregnant in the near future. What advice do you give her regarding her anti-thrombotic medication during the anticipated pregnancy?

- A. Continue with warfarin until third trimester
- B. She should replace warfarin with aspirin at analgesic dose.
- C. Discontinue all drugs that affect the blood
- D. Replace warfarin with heparin

26. What type of drugs can cross the blood-brain barrier (BBB)?
- A. Large and lipid-soluble
 - B. Large and lipid-insoluble
 - C. Small and lipid-soluble
 - D. Small and lipid-insoluble
27. If protein plasma binding is decreased, how will volume of distribution be affected?
- A. Increased
 - B. Decreased
 - C. Not changed
28. Drug biotransformation phase I makes drugs ____ polar for metabolism and phase II makes drugs ____ polar for excretion.
- A. More; More
 - B. More; Less
 - C. Less; More
 - D. Less; Less
29. Which of the following reactions is phase II and NOT phase I?
- A. Oxidations
 - B. Reductions
 - C. Conjugations
 - D. Deaminations
 - E. Hydrolyses
30. Regarding the microsomal drug metabolizing system, a patient with late stage alcoholism and liver damage would have more ETOH available due to which of the following concepts?
- A. Increased induction
 - B. Decreased induction
 - C. Increased inhibition
 - D. Decreased inhibition
31. Regarding the microsomal drug metabolizing system, a patient who is a chronic user of barbiturates would need more drug to produce the same effects due to which of the following concepts?
- A. Increased induction
 - B. Decreased induction
 - C. Increased inhibition
32. According to the biopharmaceutical classification system (BCS), drug substances having high solubility and low permeability is
- A. Class-A
 - B. Class-B
 - C. Class-C

D. class-D

33. Which route of drug administration is most affected by first pass effect?

- a. Oral
- b. Rectal
- c. Vaginal
- d. SC

34. Two drugs show the same $t_{1/2}$. They will show

- A. the same Cl
- B. The same V_d
- C. The same Elimination rate constant
- D. The same

35. Assume a drug eliminated through enzymatic metabolism. The drug's plasma concentration decreases by 4 ng/ml every 2 hours. The elimination rate constant describing this metabolism will have the unit: 1/hr

- A. 1/hr
- B. Mg/ml
- C. Mg/hr
- D. Ml/hr

36. Drug A has a half-life of 4 hours & Drug B has a half-life of 12 hours. From this we can conclude that

- A. The clearance of Drug A is greater than the clearance of Drug B.
- B. The clearance of Drug B is greater than the clearance of Drug A.
- C. The Elimination rate constant of Drug B is greater than the clearance of Drug A.
- D. The V_d of Drug B is greater than the clearance of Drug A.

37. A drug has a half-life of 4 hours. A level of 10mg/L is desired. When this drug is infused at a rate of 20mg/hr, it takes 20 hours to reach this plasma level. If the drug is infused at a rate of 40mg/hr instead, the time required to reach the desired level will be hours

- a. decreased to 4
- b. increased by 4
- c. remain the same
- d. decreased 2

38. Phase –III of the Myo electric movement of the stomach does not refers to

- a. Allows evacuation of all the contents of the stomach
- b. Pellets and Oral solution do not wait for phase III even if taken immediately after food
- c. Enables opening of the pylorus.
- d. None

39. In Bioequivalence(BE) Studies(Comparison of Brand and generic)does not include

- a. In vitro Studies is considered sufficient for Class III drugs

- b. Two products are bioequivalent if they are pharmaceutically equivalent and their bioavailabilities are similar
 - c. Similarity tests of (f2) can be used to testify similarly of generics and innovator products in vitro.
 - d. None
40. One of the following is true about Hydrates APIs
- A. Amoxicillin, the mono hydrate form is less soluble than the trihydrate form
 - B. Amoxicillin, the mono hydrate form is more bioavailable than the trihydrate form
 - C. Azithromycin the mono hydrate form is less bioavailable than the dihydrate form
 - D. All hydrate have equal solubility and bioavailability
41. In an orally administered solid dosage form containing a weak electrolyte drug, one of the following is not true about the bioavailability
- a. Dosing ketoconazole (weakly basic) 2 hours after the administration of the H₂ blocker cimetidine, results in a significantly reduced bioavailability.
 - b. Absorption of weakly basic drugs starts in the stomach where they are in unionized form.
 - c. Weakly acidic drugs (e.g. Aspirin) have increased onset of action if taken in empty stomach.
42. In bio-pharmaceutics one of the following is not true about **a 'sink'** condition
- a. Shortly after a meal, there will be an increase in a **'sink'** condition
 - b. The 'sink' conditions ensure increased passive diffusion.
 - c. Increased 'sink' condition may result in decreased metabolism of some drugs (e.g. propranolol).
 - d. a 'sink' condition is a prerequisite for drugs absorbed by carrier mediated mechanism
43. Which of the following drug permeation mechanisms involves polar substances too large to enter cells by other means, such as iron or vitamin B12?
- A. Aqueous diffusion
 - B. Lipid diffusion
 - C. Carrier molecules
 - D. Endocytosis and exocytosis
44. Which of the following is the amount of a drug absorbed per the amount administered?
- A. Bioavailability
 - B. Bioequivalence
 - C. Bioinequivalence
 - D. Dosage
45. When is consumption method of drug quantification appropriate?

- A. if there is accurate consumption data
- B. if drug supply is consistent
- C. if there is appropriate stock management system
- D. if wastage and expiry are excessive
- E. all except D

46. Which is not the disadvantage of consumption method?

- A. does not encourage morbidity recording
- B. Requires less detailed calculation
- C. not reliable if there have been long stock outs
- D. may not be appropriate for rapidly expanding service

47. Morbidity method of drugs quantification is appropriate if;

- A. available consumption data are incomplete
- B. prescribing patterns are not cost effective
- C. budget is not adequate
- D. all of the above

48. One is not part of a procurement principle

- A. procure by generic
- B. procure from essential drug list
- C. procure in smaller quantity
- D. apply transparent and written procedure

49. Which method of procurement has the list price

- A. open tender
- B. restricted tender
- C. direct procurement
- D. competitive negotiation

50. When should stock records be updated?

- A. whenever stock movement occurs
- B. during every physical inventory
- C. in cases of losses
- D. every day
- E. all except D

51. The stock level that is kept to reserve to prevent stock outs is termed as;

- A. buffer stock
- B. safety stock
- C. reorder stock level
- D. A and B

52. Which of the following is less likely to affect rational use of drugs?

- A. prescribing practice
- B. dispensing practice
- C. patient adherence
- D. good manufacturing practice (GMP)
- E. all of the above

53. The external users of accounting information are;

- a. Creditors
- b. Investors
- c. Marketing department
- d. Finance department
- e. A and B

54. A business organized as a separate legal entity is known as;

- a. Proprietorships
- b. Partnership
- c. Corporation /share company
- d. Privately limited company/PLC

55. Claims against assets are called;

- a. Liabilities
- b. Owners' equity
- c. Accounts payable
- d. A and C

56. The components of a marketing mix as a tangible good or service is ;

- a. Place
- b. Price
- c. Promotion
- d. Product

57. Placing your brand products in the minds of the customer is best explained by;

- a. Product positioning
- b. Market segmentation
- c. Direct marketing
- d. Targeting

58. One of the following is a primary data source in market research, identify;

- a. Market journals
- b. Trade publication
- c. Focus group discussion
- d. None

59. All are internal factors that affect pricing decision except,

- a. Marketing objective
- b. Marketing mix strategy
- c. Costs
- d. Competitors' price

60. The type of pricing whereby low price is imposed to attract a large number of buyers is;

- a. Penetration
- b. Market skimming
- c. Premium pricing
- d. Value pricing

61. One is not an element of porter's five force model

- a. Entry barriers
- b. Potential substitutes
- c. Political environment
- d. Competitor rivalry

62. Who is responsible to make sure that effective, safe and quality medications are available to the population?

- A. manufactures
- B. the distribute system
- C. regulatory authorities
- D. health professionals
- E. all of the above

63. All are considered as criteria for selecting medicines, except?
- A. prevalence of the diseases
 - B. efficacy, safety and quality
 - C. cost benefit ratio
 - D. branded medicine
 - E. selection using generic name
64. Which one of the following methods can be used to detect telescopic defect of filled hard gelatin capsules?
- A. Leakage test
 - B. Visual inspection
 - C. Dissolution test
 - D. Weight variation test
65. When compared to capsule manufacturing, tablet manufacturing:
- A. Require fewer unit operations
 - B. Is slower process
 - C. Has higher production rate
 - D. Needs fewer equipments
66. Quality control test that should be routinely monitored while manufacturing liquid dosage forms include:
- A. Viscosity test
 - B. pH test
 - C. Content uniformity
 - D. All
67. Which one of the following statements concerning pharmaceutical water treatment is **wrongly** stated?
- A. Sand filters are used to remove coarse particles
 - B. Ion exchange resins are used as deionizers and water softeners
 - C. Distillation avoids microbial and endotoxin contamination
 - D. USP purified water is used for the preparation of injectable drugs
68. In the treatment of pharmaceutical water, dissolved organic materials found in the incoming water can be removed by:
- A. Ion exchangers
 - B. Carbon filters
 - C. Membrane Filtration
 - D. UV light
69. EPHARM wants to incorporate one aqueous and one non aqueous preservative in to an emulsion dosage form which it is currently developing. Which one of the following could be used?
- A. Ascorbic acid & sodium metabisulphite
 - B. Ascorbic acid & sodium sulphite
 - C. Sodium bisulphite & Butylatedhydroxytoluene
 - D. Butylate dhydroxyanisol & Butylatedhydroxytoluene
70. In an emulsion preparation, equipment that is used to achieve dispersion by forcing mixture of liquids through a small inlet orifice/nozzle at high-pressure is:
- A. Homogenizer
 - B. Mechanical stirrer
 - C. Rotor-stator colloid mill
 - D. None

71. If a company wants to depyrogenate glass bottles to be filled aseptically, which one of the following methods is more preferable?
- A. Moist heat sterilization
 - B. Dry heat sterilization
 - C. Radiation Sterilization
 - D. Gaseous sterilization
72. In its formula, a parenteral formulation consists API, water for injection, citric acid, methyl paraben and sodium chloride. In this formulation, sodium chloride is used as:
- A. A tonicity adjuster
 - B. An antimicrobial preservative
 - C. A buffer
 - D. An antioxidant
73. Which of the following is NOT the scope of new clinical oriented pharmacy service?
- A. Provision of drug information for patients and health professionals
 - B. Provision of medicine use counseling to patient
 - C. Diagnosis and prescription of medications for patients
 - D. Controlling of medicines supply management
 - E. Monitoring drug therapy for safety and efficacy
74. Drug therapy problem in which the patient either requires drug therapy or taking unnecessary drug is.
- A. Unsafe drug therapy
 - B. Ineffective drug therapy
 - C. Inappropriate drug therapy
 - D. Nonadherence
75. The following laboratory findings are consistent with acute bacterial infections except:
- A. Leukocytosis
 - B. Pancytopenia
 - C. Neutrophilia
 - D. Left shift
76. Which of the following is not a common cause of peptic ulcer disease (PUD)?
- A. Alcohol ingestion
 - B. Nonsteroidal anti-inflammatory drugs (NSAIDs)
 - C. Stress-related mucosal damage
 - D. Helicobacter pylori infection
77. M.R is a 70-year-old woman suffering from arthritis requiring prolonged use of NSAID, which of the following is the preferred therapy to prevent NSAID-induced peptic ulcer disease:
- A. Misoprostol
 - B. Rofecoxib
 - C. Omeprazole
 - D. Famotidine

78. Which of the following agent is most effective for prevention of nausea and vomiting associated with motion sickness and vertigo?
- A. Scopolamine
 - B. Metoclopramide
 - C. Ondansetron
 - D. Larazepam
79. The preferred initial treatment option for a 45-year-old male presenting with a 3-month history of severe, continuous Gastroesophageal reflux disease is:
- A. Patient-directed therapy with OTC omeprazole
 - B. Prescription strength H₂-receptor antagonist
 - C. Prescription strength proton pump inhibitor
 - D. Anti-reflux therapy
80. For rapid relief of mild heart burn associated with GERD, which of the following is appropriate
- a. Proton pump inhibitors
 - b. Antacids
 - c. Histamine 2 receptor blockers
 - d. Prokinetic agents
81. Which of the following antipsychotic is preferred as first line agent?
- A. Chlorpromazine,
 - B. Haloperidol
 - C. Risperidone
 - D. Clozapine
82. O.Y. is a 27-year-old male with schizophrenia. This is his 4th psychiatric hospitalization. He has most recently been treated with risperidone. He currently presents in an acute psychotic episode with *fulminant suicidal ideation* and a serious attempt prior to hospitalization. Which of the following antipsychotics would be the best choice for this patient?
- A. Quetiapine
 - B. Clozapine
 - C. Haloperidol
 - D. Risperidone
83. Of the following combinations of medications, which one should avoid due to risk of serotonin syndrome?
- A. Fluoxetine–lithium
 - B. Fluoxetine–phenelzine
 - C. Citalopram–valproic acid
 - D. Citalopram–aripiprazole
84. Mr K.A is 62 yrs old Ethiopian, whose BP is persistently elevated to 165/98 mmHg, has known left ventricular dysfunction (systolic heart failure). On physical examination the patient was

found to have edemas in his feet and hands. What is the drug of choice for this patient at this time;

- A. Furosemide
- B. Spironolactone
- C. Enalapril
- D. Chlorthalidone

85. Rapid bronchodilation and relief acute symptoms in asthmatic asthma can be achieved by use of

- A. Salmeterol
- B. Theophylline
- C. Albuterol
- D. Fluticasone

86. Which of the following is primary long-term medication for control of moderate persistent asthma in 4-year-old child?

- A. Salmeterol twice daily
- B. Fluticasone propionate twice daily
- C. Sustained-release theophylline BID
- D. Montelukast once daily

87. RD is 21 years old asthmatic athletic women, who suffer from dyspnea and shortness of breath during exercise. As a clinical pharmacist what would you counsel this patient to prevent exercise induced asthma?

- A. She should stop exercise
- B. To take her albuterol 2 hours before exercise
- C. To use long term oral prednisone
- D. To take her inhaled corticosteroid before exercise

88. Type 2 diabetes mellitus is characterized by:

- A. Autoimmune destruction of β -cells in the pancreas
- B. Absolute deficiency insulin
- C. Presentation with acute complication like DKA
- D. A relative insulin deficiency with peripheral insulin resistance

89. Which of the following intervention is recommended prevention of nephropathy in diabetic patients with hypertension (HTN):

- e. Initiation of ACE inhibitors
- f. Use of loop diuretics for HTN
- g. Initiation of beta blockers
- h. Monthly monitoring of kidney function

90. Which of the following is false about metformin?

- A. All patients with type II DM should take it unless contraindicated
- B. Its major side effects are weight gain and hypoglycemia
- C. It reduces hyperglycemia by sensitizing insulin receptors
- D. It helps to lose weight

91. Which of the following is true about NSAIDs?

- A. They are effective analgesics for mild to moderate pain
 - B. They are recommended for inflammatory pain
 - C. They are used as opioid sparing agents in controlling severe pain
 - D. All of the above
92. The acute chemotherapy induced nausea and vomiting is mostly controlled by:
- A. Metoclopramide
 - B. Promethazine
 - C. Ondansertron
 - D. Lorazepam
93. The initial drug therapy for patients with chronic myeloid leukemia is
- A. Prednisone + vincristine + Asparaginase
 - B. Doxorubicin + cytarabine
 - C. Imatinib
 - D. Fludarabine
94. The majority of lung cancer cases are strongly associated with
- A. Cigarette smoking
 - B. High fatty diet
 - C. Physical inactivity
 - D. Genetic mutation
 - E. Activation of protooncogen
95. Preferred regimen for initiating antiretroviral therapy in treatment naïve HIV patients is
- A. TDF/3TC/NVP
 - B. ZDV/3TC/EFV
 - C. TDF/3TC/DTG
 - D. ABC/3TC/NVP
96. Which of the following **is not** the goal of ART?
- A. Improving immunologic functioning ($\uparrow$ CD4)
 - B. Prevent HIV related morbidities
 - C. Suppress plasma HIV viral load
 - D. Complete cure of HIV infection
 - E. Prevention of transmission
97. Which of the following options is the best antibiotic treatment for this patient's PCP?
- A. Bactrim IV 15 mg/kg of by TMP per day (divided intotid to qid) x 21 days
 - B. Atovaquone po 750 mg bid x 21 days
 - C. Dapsone 100 mg po daily x 21 days
 - D. Pentamidine IV 4 mg/kg per day daily x 21 days
98. Severe malaria caused by plasmodium Falciparum should be treated with
- A. Quinine HCl 600mg po TID for 7 days
 - B. Artemether 80 mg + lumefantrine 240mg po bid for 3 days
 - C. Artesunate, 2.4mg/Kg IV or IM given on admission (time = 0), then repeat at 12hours, and 24 hours, then once a day for up to 5 days.
 - D. Chloroquine 1g then 500mg in 6hr followed by 500mg QD for 2 days
 - E. None

99. Which of following is/are recommended immunization according to EPI program administered **at birth**?
- A. BCG and meningococcal vaccine
 - B. Measles and Pentavalent
 - C. BCG and OPV
 - D. Pentavalent and Rota virus vaccine
100. Malignant tumor with **no** involvement of local lymph nodes and no distant metastases is stage
- A. Stage I
 - B. Stage II
 - C. Stage III
 - D. Stage IV
101. One of the serious cancer chemotherapy toxicities, hemorrhagic cystitis associated with cyclophosphamide or ifosfamide can be prevented with:
- A. Azathioprine
 - B. Mesna
 - C. Prednisone
 - D. Aspirin
102. The use of systemic **therapy before surgery** to make tumor operable by reducing tumor burden and destroy micro-metastases is
- A. Adjuvant chemotherapy
 - B. Neoadjuvant chemotherapy
 - C. Targeted therapy
 - D. Local therapy
103. The best treatment of opioid induced constipation is:
- A. Prevention with the proper intake of fluids, fiber, and stimulant laxatives
 - B. Prevention by using propoxyphene
 - C. Concomitant use of acetaminophen with the opioid
 - D. Concomitant use of aspirin
 - E. All the above
104. Which of the following would be the best regimen to initiate in a diabetic, hypertensive patient with stage 4 chronic kidney diseases (eGFR<30ml/min)?
- A. ACEI plus thiazide diuretic
 - B. ACEI plus loop diuretic
 - C. CCB plus β -blocker
 - D. ACEI plus β -blocker
105. SR is 23 yrs old woman had unsafe sex with her boyfriend last night. Now she came to your pharmacy to avoid unintended pregnancy. Which of the following is appropriate drug for her:
- A. Yasmin (EE 30mg + drospirenone 3mg) daily for 21 days
 - B. Levora (EE 30 mcg Levonorgestrel 0.15 mg) daily continuously
 - C. Two tablets 0.75 mg of levonorgestrel take 12hrs apart
 - D. Depot MPA 150 mg Injection

- E. All can be used
106. The anemia which is characterized by microcytosis and hypochromia is
- A. Megaloblastic anemia
 - B. Iron deficiency anemia
 - C. Hemolytic anemia
 - D. vitamin B12 deficiency anemia
107. Which of the following antiarrhythmic drugs are used for treatment of atrial fibrillation acting by slowing ventricular rate (rate control)?
- A. Digoxin
 - B. Beta blockers
 - C. Calcium Channel Blockers
 - D. All of the above
108. Which of the following antidepressant **least likely** cause sexual dysfunction:
- A. Amitriptyline
 - B. Bupropion
 - C. Fluoxetine
 - D. Venlafaxine
109. Which of the following should be done to appropriately select an antimicrobial regimen?
- A. Confirm the presence of an infection
 - B. Identify the pathogen causing the infection
 - C. Select the antimicrobial regimen based on the site of infection
 - D. Monitor the therapeutic response
 - E. All of the above

110. In the absence of a complicating bacterial infection, which of the following is the most appropriate approach to treating acute bronchitis?

- A. Prescribing broad spectrum antibiotics
- B. Routinely using nonprescription cough & cold preparations.
- C. Providing symptomatic and supportive care
- D. Discouraging hydration and bed rest

111. Which drug regimen you would be recommended to treat newly diagnosed TB?

A. INH, RIF, PZA x 8 weeks, then INH, RIF x 18 weeks

B. INH x 9 months

C. INH + RIF x 9 months

D. INH + RIF + EMB + FQ x 8 wks, then INH + RIF x 18 wks

E. INH + RIF + EMB + PZA x 8 wks, then INH + RIF x 18 wks

112. What is the role of Pyridoxine (VitB6) for alcoholic, pregnant and HIV positive patients on anti- TB treatment?

- A. Prevention of hepatotoxicity associated with Rifampin
- B. Prevention of peripheral neuropathy associated with isoniazid
- C. Treatment of retinopathy associated with Ethambutol
- D. Treatment of gout associated with Pyrazinamide

113. A 44-year-old obese male has a significantly high level of plasma triglycerides. Following treatment with one of the following agents, his plasma triglyceride levels decrease significantly to almost normal. By which of the following agents is it possible to get this reduction?

- a. Neomycin
- b. Cholestyramine
- c. Lovastatin
- d. Gemfibrozil

114. A 9-year-old child is having learning difficulties at school. He has brief lapses of awareness with eyelid fluttering that occur every 5-10 min. EEG studies reveal brief 3-Hz spike and wave discharges appearing synchronously in all leads. Which drug would be effective in this child without the disadvantages of excessive sedation or tolerance development?

- A. Clonazepam
- B. Diazepam

- C. Ethosuximide
 - D. Phenytoin
 - E. Phenobarbital
115. Mr. Y is a 35 year old who came to your clinic and treated with ampicillin 500mg qid for 10 days. Six days later, he presents with severe skin rash as a result of ampicillin and you wish to change his medication. Which drug is most likely will not be your preferred option?
- A. Gentamicin
 - B. Clindamycin
 - C. Ceftriaxone
 - D. Erythromycin
116. A patient with a *P. aeruginosa* infection is receiving intravenous gentamicin. The aminoglycoside blood levels are well above the minimum inhibitory concentration (MIC), but the clinical response is not satisfactory. A new medication order calls for adding penicillin, administered in a separate IV line to avoid a physical incompatibility. If this order is carried out, which of the following is most likely to occur?
- A. The patient is likely to develop *Clostridium difficile* colitis
 - B. The penicillin will act synergistically with the aminoglycoside
 - C. The penicillin will decrease the risk of aminoglycoside nephrotoxicity
 - D. The risk of inducing resistance to both drugs increases dramatically
117. A patient with AIDS is treated with a combination of agents, one of which is zidovudine. Which of the following enzymes or replicative processes is the main target of this antiviral drug? Reverse transcriptase
- A. RNA synthesis
 - B. Viral particle assembly
 - C. Viral proteases
118. Mr. X, who is an accountant, is HIV positive and is on multidrug ARV regimen. After taking his medication for one month he complained that he is unable to sleep and concentrate on his routine activities. The most probable medication which is responsible for this untoward effect is:
- A. Nevirapine
 - B. Zidovudine
 - C. Efavirenz
 - D. Indinavir
119. A 20-year-old Pharmacy student is given a prescription for asthma, which the physician states has a very high therapeutic index. Which of the statements best characterizes the drug as it relates to the therapeutic index?
- a. The drug's serum levels will likely need to be carefully monitored.
 - b. The drug is likely to cross the blood-brain barrier.
 - c. The drug is likely to have extensive drug-drug interactions.
 - d. The drug is unlikely to have any serious adverse effects

- 120.** Adera is a 70-year-old asthmatic man comes in for a checkup and complains that he is having some difficulty in starting to urinate. Physical examination indicates that the man has a blood pressure of 165/105 mm Hg and a slightly enlarged prostate. Which of the following medications might be responsible for the occurrence of the two conditions?
- a. Isoprenaline
 - b. Prazosin
 - c. Timolol
 - d. Phentolamine
- 121.** A friend has very severe hypertension and asks about a drug her doctor wishes to prescribe. Her physician has explained that this drug is associated with tachycardia and fluid retention (which may be marked) and increased hair growth. Which of the following is most likely to produce the effects that your friend has described?
- A. Captopril
 - B. Guanethidine
 - C. Minoxidil
 - D. Propranolol
- 122.** A 55-year-old patient has been referred to you. She complains about a skin rash and a cough. In the course of history taking, she tells you that she takes high blood pressure medication but she doesn't remember the name. You suspect drug toxicity. Which of the following antihypertensive agents is the patient most likely taking?
- a. Captopril
 - b. Nifedipine
 - c. Prazosin
 - d. losartan
- 123.** 123.A 30 year old man is recently diagnosed with HIV and therapy is started. After the first week of therapy, the patient complains of headaches, irritability, and nightmares. Which one of the following anti-retroviral drugs is most likely to be causing these symptoms?
- A. Efavirenz
 - B. Indinavir
 - C. Lamivudine
 - D. Nevirapine
 - E. Stavudine
- 124.** 124.A quality control company wants to inspect the manufacturing quality of the drug manufacturer. What the inspector inspects?
- A. Good laboratory practice
 - B. Good dispensing practice
 - C. Good manufacturing practice
 - D. Good procurement practice
- 125.** You are working in the drug manufacturing company and prepares drugs of having trace metals and air sensitive formulation. Which of the following excipient(s) are used to formulate this drugs?
- A. Chelating agent and preservative
 - B. Chelating agent and anti-oxidant
 - C. Chelating agent

D. Preservative

- 126.** 126.A company wants to register the drug manufactured. What is the period product submission for registration to the drug quality control (e.g. EFMHACA)
- a. 120days
 - b. 150days
 - c. 30days
 - d. 180days
 - e. None
- 127.** 127.If the drug registration period of submission was past. When is the minimum period for reregistration?
- A. 30days
 - B. 15days
 - C. 120days
 - D. 60day
- 128.** 128.A quality control company wants to inspect the manufacturing quality of the drug manufacturer. What the inspector inspects?
- a. Good laboratory practice
 - b. Good dispensing practice
 - c. Good manufacturing practice
 - d. Good procurement practice
- 129.** A pharmacist wants to prepare two incompatible active ingredients in one tablet. Which of the following methods are used for this preparation?
- A. Press coating
 - B. Enteric coating
 - C. Sugar coating
 - D. Sustained release preparation
- 130.** A pharmacist wants to prepare paste to his client. After preparation the pharmacist recommend the client to apply on the skin. During application the paste does not create cooling effect to the skin. What component is the paste lacks?
- 131.** Which methods of drug manufacturing is used when the drugs are moisture sensitive and sensitive with high temperature?
- A. Dry granulation
 - B. Wet granulation
 - C. Pulverization
- 132.** If the in quality control officer wants to test the quality of the bulk and finished product of a drug. If finished product quality control test is failed, what should he do?

- 133.** A pharmaceutical analyst wants test a drug by measuring the absorbance of a drug by analytical instrument. Which of the following analytical instrument is used for this measurement?
- A. Dissolution tester
 - B. UV spectrophotometers
 - C. Disintegration tester
 - D. HPLC
- 134.** One of the following is not true about a Pharmacoeconomics evaluation.
- a. Concerns in economic evaluation are always to select a minimum cost strategy.
 - b. Increase in health care expenditures necessitated the demand for economic evaluation.
 - c. Threshold values help to decide whether a strategy is cost effective or not.
 - d. None
- 135.** Which consequences of a medical care are most useful in economic analysis?
- a. Humanistic
 - b. Economic
 - c. clinical
 - d. all are equally important
- 136.** The outcome that considers either safety or efficacy end points of a disease treatment is?
- a. Humanistic
 - b. Economic
 - c. clinical
 - d. none
- 137.** Which of the following is a desired (positive) outcome?
- a. Number of epidemic cases cured
 - b. Reduction in HbA1c level among type II diabetes patients
 - c. Number of adverse drug reactions from a treatment
 - d. Number of deaths from an operation
 - e. a & b
- 138.** One of the following can be regarded as an intermediate outcome in a treatment.
- a. Low cholesterol levels in a myocardial infarction
 - b. Reduction in HbA1c from type II diabetes mellitus
 - c. Reduction in Blood pressure levels from hypertension
 - d. All of the above
 - e. None of the above

- 139.** Which one of the following is associated with the result of a drug in realistic conditions?
- Efficacy
 - Effectiveness
 - efficiency
 - equity
- 140.** Which of the following could not be the fundamental reason for the need of a Pharmacoeconomics evaluation?
- Raise of healthcare costs against restricted resources
 - Decision making about affordability and accessibility of drugs would be simplified
 - Need for evidence about claims of a manufacturer on its new drugs
 - Helps to investigate the mechanism of action for a drug
- 141.** Which of the following is a full economic evaluation method?
- Cost of illness
 - Cost consequence analysis
 - cost effectiveness analysis
 - all of the above
- 142.** Type of perspectives in a Pharmacoeconomics evaluation include the following except:
- Patient
 - third party/ payer
 - hospital
 - employer/society
 - none
- 143.** A treatment outcome measure that combines both quantity and quality of life is:
- QALY
 - Life years saved
 - morbidity
 - mortality
- 144.** An economic evaluation method which incorporates patients' preferences is;
- Cost utility analysis
 - Cost effectiveness analysis
 - cost benefit analysis
 - cost minimization
- 145.** When do we use a cost utility analysis (CUA)?
- If health related quality of life is an important outcome.
 - If Effectiveness data shows that outcomes are equivalent
 - If program affects wide range of outcomes
 - Program shows dominance
 - a & c are answers
- 146.** Applications of a Pharmacoeconomics evaluation would be the following except;
- Clinical decision
 - Formulary management
 - drug use guidelines
 - disease management
 - none
- 147.** At which of the drug development stage are pharmacoeconomic studies most important?
- Phase I
 - Phase II
 - Phase III
 - Phase IV
 - e/all
- 148.** If two drugs A & B are presented in such a way that drug A is more effective and equal in cost compared to drug B, which of the following would the pharmacist recommend?
- Prefer drug A over drug B
 - Prefer drug B over drug A
 - drug B is dominant over drug A
 - Drug A is dominated by drug B

- 149.** A method that helps a decision maker to examine the robustness of a decision by systematically changing certain parameters is called;
- Decision analysis
 - Sensitivity analysis
 - critical analysis
 - decision tree
- 150.** The node in a decision tree at which several options are possible and the decision maker controls which to choose is;
- Chance node
 - Decision node
 - terminal node
 - all
- 151.** Which statement is true about discounting?
- The higher the discount rate, the higher an individual prefers current consumption
 - Discounting helps to bring future costs and consequences to their present value
 - Discounting is a must when differential timing occurs in costs and consequences
 - All are true
- 152.** One of the following is not considered when critiquing a pharmacoeconomic evaluation?
- Whether all relevant costs are included
 - Whether all relevant outcomes are included
 - Sensitivity analysis
 - Type of evaluation done
 - none
- 153.** One of the economic evaluation method which applies monetary units to costs and outcomes is;
- Cost utility analysis
 - Cost effectiveness analysis
 - cost minimization analysis
 - none of the above
- 154.** Which of the following terms is related to economics?
- scarcity of resources
 - unlimited wants
 - choices
 - optimization
 - all
- 155.** Which is the basic characteristic of public goods?
- non excludability
 - non rival nature
 - non- rejectability
 - all
 - none
- 156.** Which of the following factors contribute to market failure?
- negative externality
 - abuse of market power
 - imperfect information
 - unregulated public goods
 - all
- 157.** Which methods are employed to minimize market failure?
- state provision
 - property rights
 - regulation
 - prohibition
 - a & b
 - all
- 158.** Which factor shifts the supply curve to the right?
- decrease in price of inputs
 - increase in expected price
 - decrease in number of firms
 - increase in price of inputs
- 159.** Which of the following factors shifts the demand curve to the left?
- increase in price of a substitute
 - increase in population
 - increase in income
 - increase in expected price
 - none
- 160.** The market state where both quantity demanded and supplied are equal is called;
- equilibrium point
 - excess supply

b/ equimarginal utility

d/excess demand

161. At a given price, the excess of quantity supplied over the quantity demanded is called;

a/ surplus

c/ exploitation

b/ shortage

d/ excess demand

162. Which of the following is a characteristic of the health sector?

a/ large components of personal service

c/ uneven incidence of diseases

b/ restrictions on competition

d/ consumers ignorance

e/ all

163. Which of the following statements is true?

a/ demand for primary healthcare is price inelastic.

b/ consumers are more price elastic as out of pocket payment increases.

c/ co-insurance leads for increase in demand for healthcare services.

d/ factors other than price cause the demand or supply curve to shift either to the left or right.

164. Assume that Tikur Anbessa Comprehensive Specialized Hospital is producing sterile pharmaceutical products (parenteral nutrition). If the hospital wants a simple, cheap, highly specific method for performing microbial endotoxin test, which one of the following do you recommend?

A. Sterility test

C. LAL test

B. Rabbit test

D. Particulate matter evaluation

165. The degree of sterility accepted in the testing of sterile pharmaceutical products is:

A. 99% sterile

C. 80-120% sterile

B. 95% sterile

D. None

166. Which one of the following parameters should be controlled in the cleanroom of sterile production area?

A. concentration of airborne particles

C. air pressure are controlled

B. temperature,

D. All

167. Which one of the following size reduction equipment employs both impact and attrition mechanisms?

A. Hammer mill

C. Cutter mill

B. Ball mill

D. Compression mill

168. If an operator in a pharmaceutical industry wants to dry a suspension, which one of the following is preferable?

A. Tunnel dryers

C. Hot oven drier

B. Spray drier

D. Rotary dryer

169. Advantages to the manufacturer for tablet film coating when compared to sugar coating include :

a. Shorter production time

c. Enable putting of logo on tables

b. Adaptable for controlled release

d. All

170. Which of the following is **not** used primarily as a diluent in tablet formulations?
- A. Magnesium stearate
 - B. Lactose
 - C. Starch
 - D. Dicalcium phosphate
171. A pharmacist fills 250mg of a drug into number 2 capsules and finds that the capsule is too small to accommodate the powder. She may select to use:
- A. A number 4 capsule
 - B. A number 0 capsule
 - C. A number 3 capsule
 - D. A number 5 capsule
172. **Excipients** used in the manufacturing of hard gel capsules filled with powder drug and their primary functions are given. Choose the one which is **mismatched**.
- A. Lactose- binder
 - B. Magnesium stearate- lubricant
 - C. Sodium starch glycolate- disintegrant
 - D. Sodium lauryl sulphate- surfactant
173. The **correct order** in the manufacturing of filled hard gel capsules is:
- A. Filling- capping-ejection-rectification
 - B. Capping- rectification- filling-ejection
 - C. Rectification-filling- capping-ejection
 - D. Filling-ejection-rectification-capping
174. A **tablet** coating defect in which the film cracks on the crown of the surface or splits on the tablet edge could be prevented by:
- A. Increasing plasticizer concentration
 - B. Increasing drying time
 - C. Using weak polymers
 - D. Using pigments instead of dyes
175. Which of the following statements is **true** about sterilization methods?
- A. Dry heat is more effective in killing microorganisms than steam
 - B. Steam sterilization is useful in sterilizing thermolabile drugs
 - C. Radiation sterilization is useful in sterilizing heat sensitive products.
 - D. Ethylene oxide is a non toxic gas used for sterilization of parenteral products
- A. Moist heat sterilization
 - B. Dry heat sterilization
 - C. Lyophilization
 - D. None
176. According to the **international** organization for standardization(ISO) designation of clean room, the most clean area is:
- A. Class 100
 - B. Class 1000
 - C. Class 10,000
 - D. Class 100,000

177. The **advantages of sustained release** dosage forms include all of the following **except**:

- A. Offers less frequent drug administration
- B. Offers more flexibility in adjusting dosage regimens
- C. Reduce fluctuation in steady-state level
- D. Increase safety margin of high potency drugs

178. **Which one of the** following types of preservatives is most widely used for oral liquid preparations?

- A. Mercurial preservatives
- B. Neutral preservatives
- C. Acidic preservatives
- D. Quaternary ammonium compounds

179. Validation should be necessary when:
- A. Any new master formula is adopted
 - B. New method of preparation is adopted
 - C. New equipment is used
 - D. All
180. Which one of the following statement is **correct** about GMP?
- a. GMP is a part of QC aimed at ensuring that products are consistently manufactured to a quality appropriate to their intended use
 - b. QA is laboratory based process
 - c. All containers and equipment should bear labels
 - d. Testing is part of GMP, and alone it provides a good level of quality assurance
181. SSP a new pharmaceutical company wants to manufacture Paracetamol tablet (500mg) using 100kg Paracetamol active ingredient. After the granulation and dry mixing total impute of all material was 120 kg. What should be the average weight of paracetamol for compressions?
- A. 500mg B. 600mg C. 700mg D. None
182. Evaluation of empty gelatin capsule does not include
- A. Moisture content test
 - B. Disintegration test
 - C. Wight lose on drying
 - D. General appearance
 - E. Dissolution test
183. Quality control of filled capsules does not include
- A. Weight variation
 - B. Content uniformity test
 - C. Disintegration test
 - D. Dissolution test
 - E. Friability test
184. As major differences between Sugar and Film one of the following is not true, in film coating
- A. You can coat any shape of tablet.
 - B. Pelletization is more easier than in sugar coating
 - C. automation more adaptable than sugar coating does
 - D. Control variables are more subjective than sugar coating
 - E. None of the above
- 185 . The rationales for the preference of soft gels are
- A. improved drug absorption
 - B. increased patient compliance
 - C. adaptability for oily drugs filling
 - D. Enhanced accuracy
 - E. All
186. Highly Purified Water (HPW) may not

- A. Be prepared from potable water source
- B. have same quality standard as WFI including limit for endotoxins
- C. be prepared by combination of methods including reverse osmosis (RO), ultrafiltration (UF) and deionization (DI)
- D. be a specification in the European Pharmacopoeia

187 . In dry granulation,

- A. Powders are mixed with binder solution that will result in an increase in
- B. Roller compactors are used to squeeze the powder mass which can be crumbled into "slugs," so this is called "slugging."
- C. Heavy duty compression machines can be used to compress the mix and to form big tablets which will be milled and screened latter.
- D. All of the above

188.The application of fluidized bed system/ equipment/ in pharmacy does not includes

- A. for granulation purpose
- B. for coating purpose
- C. for pelletization purpose
- D. for drying of wet solids
- E. for filling purpose

189.One of the following is not true about spray dryers

- A. used for drying of wet solids
- B. it is a Rapid drying operation
- C. It results in Products of high bulk density and raised dissolution
- D. Dried materials are hygroscopic

190. Mixers designed for semi solids

- A. Works mainly by convection and diffusion
- B. Can be a planetary mixer with narrow clearance between the paddle and the vessel
- C. Can be a turbine mixer
- D. Can be a sigma blade mixer with wider clearance between the blades and the vessel
- E. All of the above

191 . The reasons for reducing a size of a material includes

- A. To enhance handling of bulky materials
- B. To avoid segregation of powders
- C. To increase bioavailability
- D. To decrease drying time of wet masses
- E. All